

Predicting Sooty Grouse Breeding and Rearing Season Habitat Suitability in Oregon Using a Random Forest Species Distribution Model

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Abstract

Knowledge of Sooty Grouse (*Dendragapus fuliginosus*) habitat association is needed to inform conservation efforts and habitat management across Oregon's western and central regions, but fine-scale distribution for the species is limited. I used a Random Forest species distribution model trained on 2,690 occurrence records from Global Biodiversity Information Facility (GBIF) and 10 environmental predictors, spanning terrain, land cover, and climate, to predict habitat suitability across Oregon during the April–September breeding and brood-rearing period. Predictors were screened for multicollinearity using Spearman correlation and variance inflation factor analysis, and model performance was evaluated via spatial block cross-validation to account for spatial autocorrelation in occurrence data. The model achieved a mean AUC-ROC of 0.8878 ± 0.0077 across 5 cross-validation folds, indicating good discriminative performance. Total precipitation and elevation emerged as the strongest predictors of habitat suitability, consistent with Sooty Grouse's known association with montane conifer forest and elevational habitat shifts. The resulting habitat suitability and potential distribution maps identify areas of high predicted suitability concentrated in mid- to high-elevation forest across Oregon's Cascade and Coast ranges, providing a baseline for future survey design and habitat conservation planning.

Introduction

The Sooty Grouse is an omnivorous galliform that lives in forests located in the mountainous areas of the west coast of North America, stretching from Alaska to the Sierra Nevada. While almost all Sooty Grouse populations dwell in conifer rich forests during the winter months, their breeding and brood-rearing habitats range from coastal rainforest to subalpine and alpine forest (Zwickel & Bendell, 2024). The grouse usually breed and rear their broods at high elevation from the months of April to September, then migrate down to lower elevation conifer forests for the winter. Proximity to conifer forests to take refuge in during the winter seems to be a constraining factor for the species distribution, meaning it is still an important factor when considering the breeding and rearing season habitats (Zwickel & Bendell, 2024).

Species Distribution Models (SDMs) link species occurrence data to environmental predictors to characterize and predict habitat suitability across an area of interest. Machine learning techniques, such as Random Forest, are growing in popularity for this use case because of their ability to capture non-linear thresholds and mitigate multicollinearity better than traditional

linear models. Random Forest Models have been used with success for characterizing habitat associations and informing management planning, such as with the Dusky Grouse in Montana (Leipold et al., 2024) and the Greater Sage-Grouse in Utah (Picardi et al., 2020).

This project applies a Random Forest SDM to Sooty Grouse occurrence data across Oregon, using terrain, land cover, and climate predictors derived from Google Earth Engine, to (1) identify the environmental variables most strongly associated with Sooty Grouse breeding and brood-rearing season habitat suitability, and (2) produce spatially explicit habitat suitability and potential distribution maps that can serve as a baseline for future survey and conservation planning efforts in Oregon.

Methods

Occurrence Data. Sooty Grouse occurrence records were retrieved from the Global Biodiversity Information Facility (GBIF) using the occurrence search API. Records were filtered to only data from after 2008, the breeding and brood-rearing season (April–September), and deduplicated to one occurrence per one kilometer raster pixel.

Area of Interest. The study area encompassed a sub-section the state of Oregon, USA, clipped both to the state boundaries of Oregon and a bounding box with a 20 kilometer buffer around the Sooty Grouse occurrence points. Environmental predictors and model outputs were generated at a spatial resolution of 1 km.

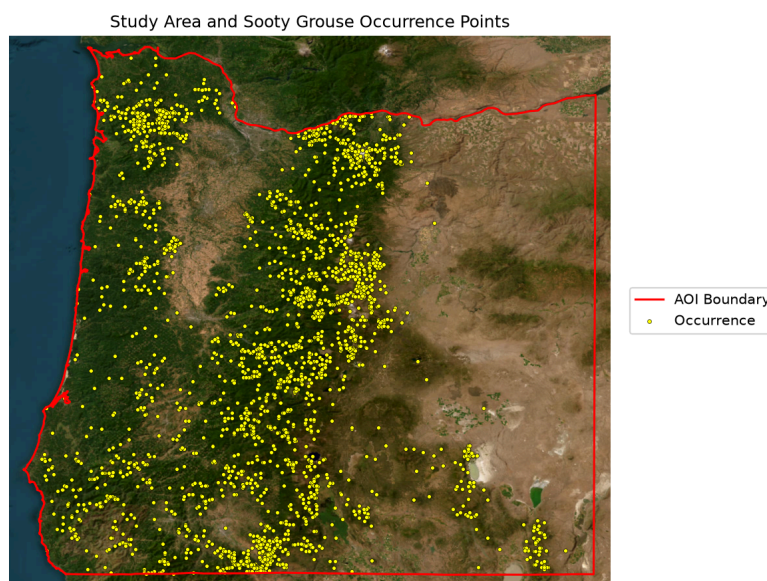


Figure 1. Study area and Sooty Grouse occurrence points used for training and evaluation, Oregon, USA.

Environmental Predictors. Ten environmental predictors were derived from three data sources within Google Earth Engine. This follows the structure of Google Earth Engine's community species distribution modeling tutorial (Yu, 2024), itself an adaptation of the SDM workflow originally developed by Crego et al. (2022).

- Terrain (NASA SRTM Digital Elevation Model, USGS/SRTMGL1_003): elevation, slope, and aspect. Aspect was transformed into northness and eastness components (cosine and sine of aspect in radians, respectively) to avoid the circularity inherent in a 0–360° directional variable.
- Land cover (USGS National Land Cover Database, 2021 release): fractional cover of evergreen forest, mixed forest, shrub/scrub, and herbaceous vegetation classes, computed as the mean of 30 m binary class rasters aggregated to the model's 1 km working resolution.
- Climate (WorldClim v1 BIO variables): mean temperature and total precipitation of the breeding/rearing season (April–September), selected in place of annual climate normals to match the temporal scope of the occurrence data.

All predictors were resampled to a common 1 km resolution and coordinate reference system (EPSG:4326) prior to model fitting. Terrain and land cover predictors, native to 30 m resolution, were aggregated using a mean reducer before reprojection to avoid the spatial aliasing that results from naive resampling of fine-resolution categorical and continuous data to a coarser grid. Climate predictors, native to approximately 1 km resolution, required no aggregation and were reprojected directly. Below are mappings (Figures 2, 3, and 4) of one predictor from each of the three data sources.

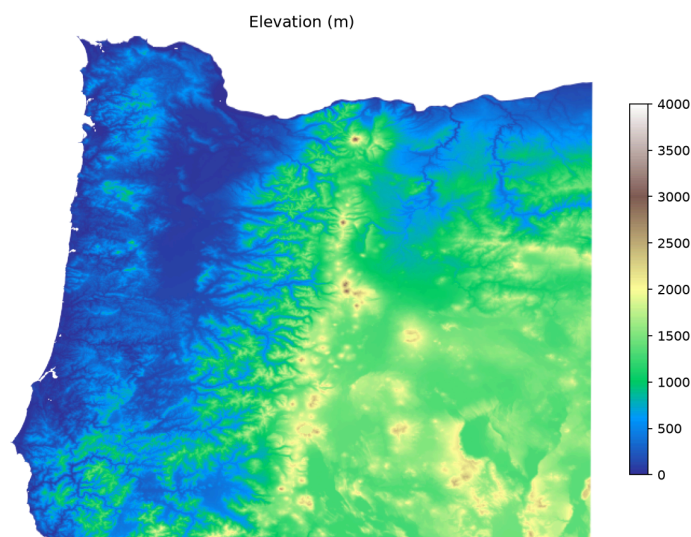


Figure 2. Elevation (meters) map of the area of interest

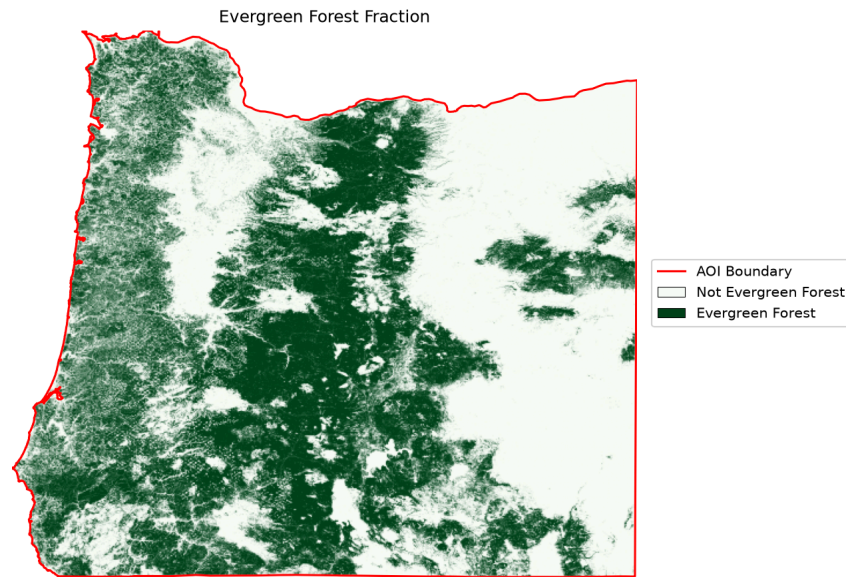


Figure 3. Evergreen Forest fraction in the area of interest

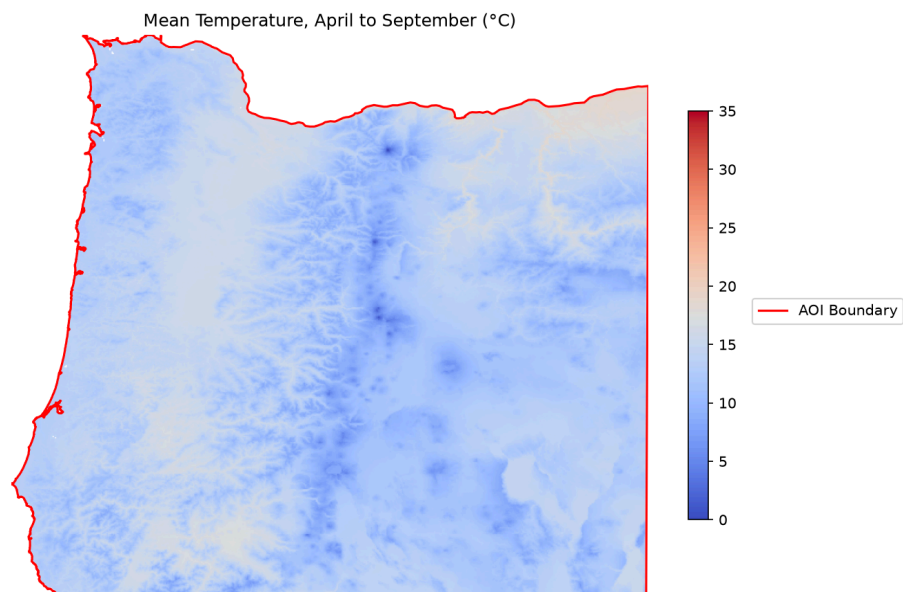


Figure 4. Mean temperature (°C) from April to September in the area of interest

Predictor Screening. Predictors were screened for multicollinearity in two stages. First, pairwise Spearman rank correlations were computed among all ten predictors, following an approach consistent with prior grouse habitat modeling studies (e.g., Leipold et al., 2024).

Second, Variance Inflation Factor (VIF) was calculated iteratively, removing the predictor with the highest VIF above a threshold of 10 and recalculating until all remaining predictors fell below the threshold.

Pseudo-absence Generation. Pseudo-absence points were sampled from a buffer-exclusion area (Figure 5). A 5 km buffer was drawn around each occurrence point, and the remaining eligible area was used as the sampling region for pseudo-absence points. The eligible area was also constrained to land area using an elevation-derived water mask.

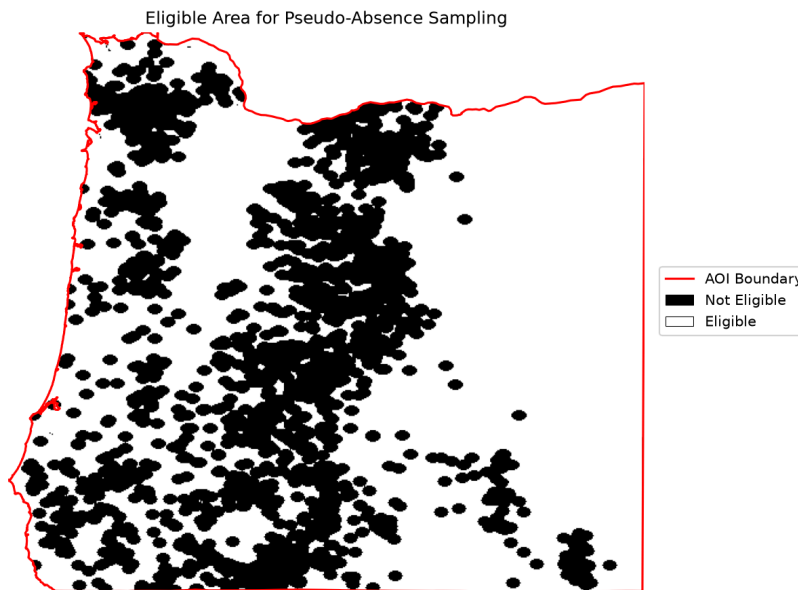


Figure 5. Map of the eligible area for pseudo-absence sampling after buffering around occurrence points.

Model. Random Forest (Breiman, 2001) is an ensemble machine learning algorithm that aggregates predictions across many decision trees, reducing overfitting relative to any single tree while handling correlated and non-linear predictors well.

A Random Forest classifier (300 trees, minimum leaf population of 10, bag fraction of 0.5) was trained once per fold on the fold's training partition. Rather than training separate classifiers for each output type, a single trained classifier was configured via output mode switching to produce both a continuous habitat suitability probability and a binary presence/absence classification from the same underlying decision trees. For each of the 5 cross-validation folds, the resulting fold-level classified images were cached as Earth Engine assets to avoid repeatedly re-deriving computationally expensive classifier outputs.

Cross-validation. Model performance was evaluated using spatial block cross-validation rather than random point-based splitting, to reduce bias from spatial autocorrelation among nearby occurrence and pseudo-absence points. The study area was divided into 40 km $\times$ 40 km grid blocks (Figure 6); for each of 5 cross-validation folds, blocks were randomly assigned to training (70%) or testing (30%) partitions using a fixed random seed per fold, and presence and pseudo-absence points were partitioned according to their containing block.

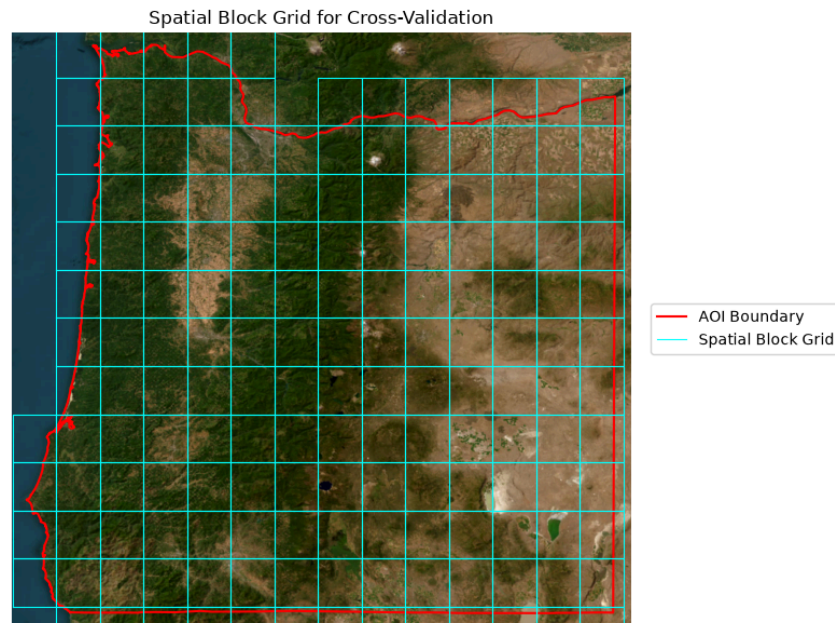


Figure 6. Spatial block grid (40 $\times$ 40 km) used for spatial block cross-validation.

Model Evaluation. For each fold, model performance was assessed using the area under the receiver operating characteristic curve (AUC-ROC) and area under the precision-recall curve (AUC-PR), computed by sampling predicted probabilities against the held-out testing partition across 25 evenly spaced probability thresholds. Final habitat suitability and potential distribution maps were generated by averaging (mean) and taking the majority classification (mode), respectively, of each fold's output image across all 5 folds.

Results

Occurrence Data. After filtering to the April–September breeding and brood-rearing season and removing duplicate records within the same 1 km grid cell, 2,690 Sooty Grouse occurrence records were retained for model training and evaluation, distributed across the Oregon study area.

Predictor Screening. Pairwise Spearman correlations among the ten candidate predictors ranged from -0.73 to 0.55 (Figure 7). The strongest correlations were observed between elevation and mean breeding-season temperature ($r = -0.73$) and between elevation and mixed forest cover ($r = -0.60$), consistent with expected relationships between elevation, climate, and forest composition in montane systems.

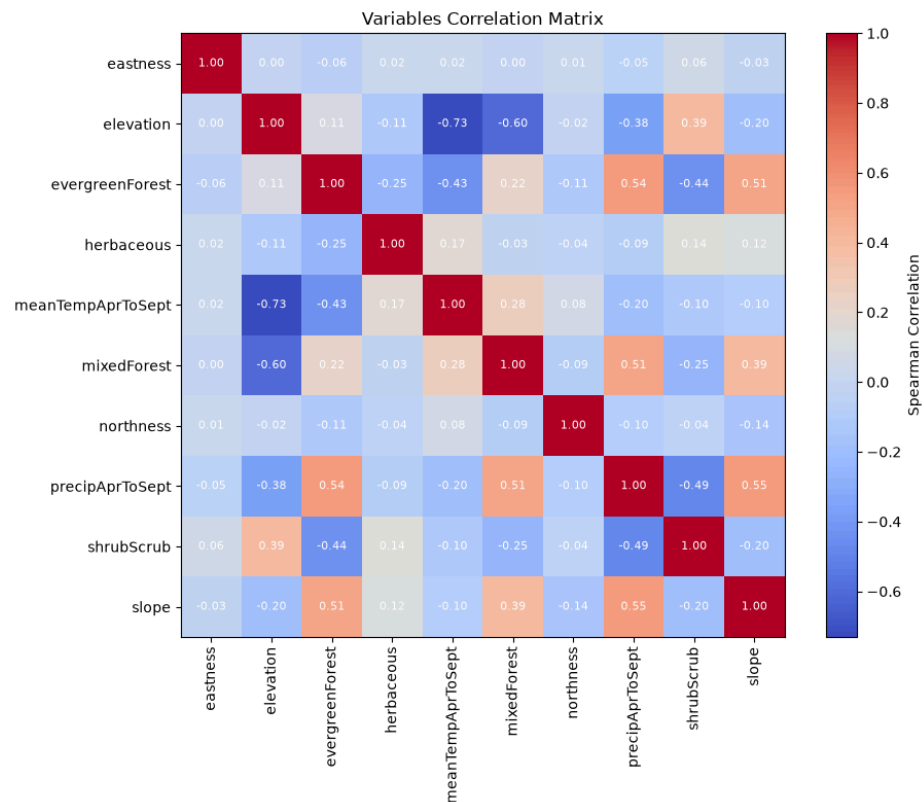


Figure 7. Spearman rank correlation matrix among the ten candidate environmental predictors.

Variance Inflation Factor (VIF) values ranged from 1.01 to 8.77 (Table 1), with elevation showing the highest value. All ten predictors were retained below or near the conventional VIF threshold of 10.

Predictor	VIF
Elevation	8.773514
Mean Temperature April to September	7.047920
Total Precipitation April to September	3.797241

Evergreen Forest Fraction	3.463953
Shrub Scrub Fraction	2.595745
Slope	1.839348
Herbaceous Fraction	1.706010
Mixed Forest Fraction	1.505195
Northness	1.041117
Eastness	1.007304

Table 1. Environmental predictors and their associated VIF value.

Model Performance. Across 5 spatial cross-validation folds, the model achieved a mean AUC-ROC of 0.888 ± 0.008 and a mean AUC-PR of 0.852 ± 0.023 (Table 2), indicating good and consistent discriminative ability across independently held-out spatial blocks.

Cross-Validation Fold	AUC-ROC	AUC-PR
1	0.886	0.829
2	0.883	0.840
3	0.899	0.882
4	0.880	0.870
5	0.891	0.837
Average	0.888 ± 0.008	0.852 ± 0.023

Table 2. Each model iteration's AUC-ROC and AUC-PR values and their averages.

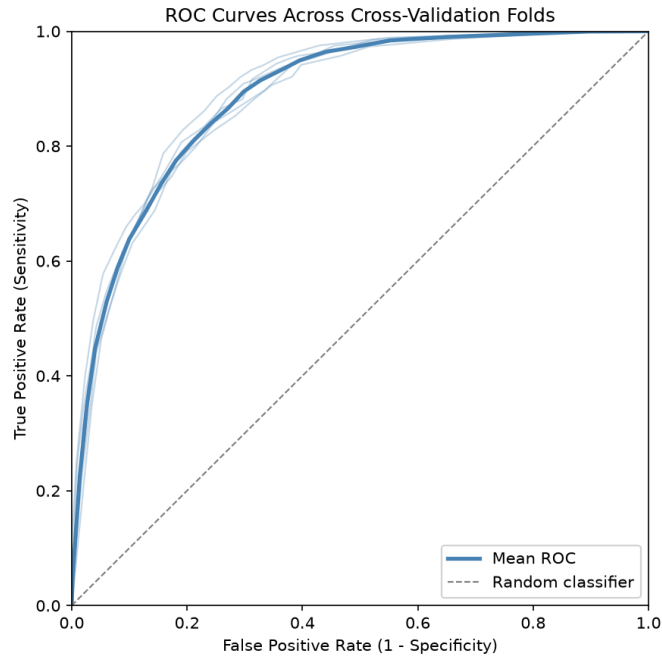


Figure 8. Receiver operating characteristic (ROC) curves for each of 5 cross-validation folds (thin lines) and the mean curve (bold line).

Variable Importance. Total precipitation and elevation were the most important predictors of Sooty Grouse habitat suitability, followed by mean temperature and slope (Figure 9). Eastness, northness, and mixed forest fraction contributed comparatively less to model predictions.

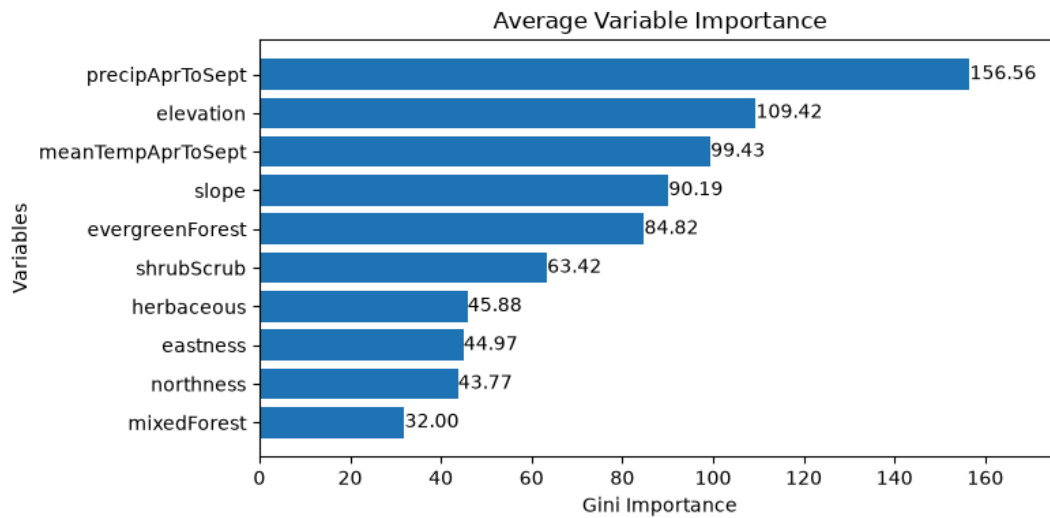


Figure 9. Mean Random Forest variable importance across 5 cross-validation folds.

Habitat Suitability and Potential Distribution. The habitat suitability map (Figure 10) shows the highest predicted suitability concentrated in mid- to high-elevation conifer forest across the Cascade Range and, to a lesser extent, the Coast Range, with low suitability predicted across the lower-elevation, drier eastern portion of the state. The potential distribution map (Figure 11), derived from majority-vote binary classification across the 5 folds, identifies the regions of the study area as predicted suitable habitat, broadly consistent with the spatial pattern of observed occurrence records.

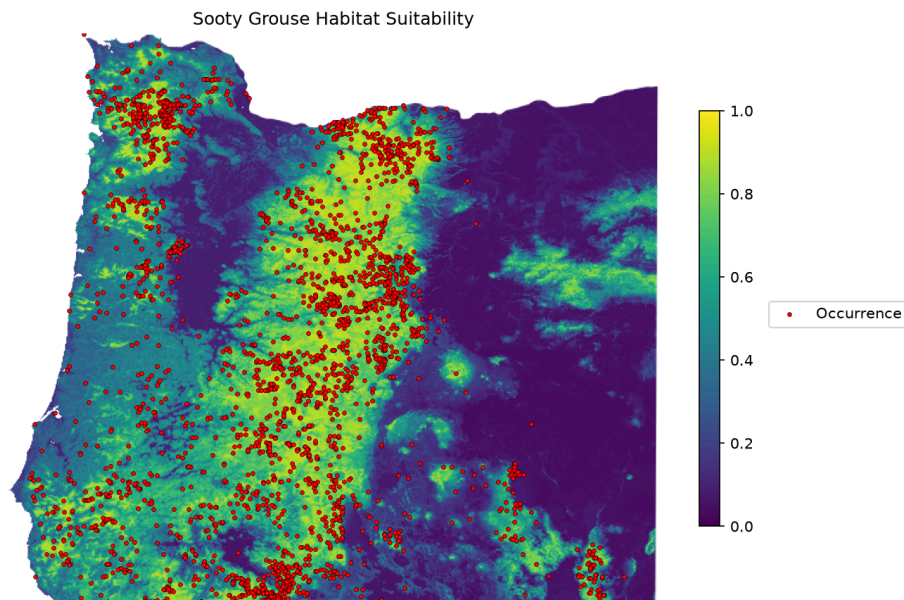


Figure 10. Predicted Sooty Grouse habitat suitability across the area of interest, averaged across 5 cross-validation folds.

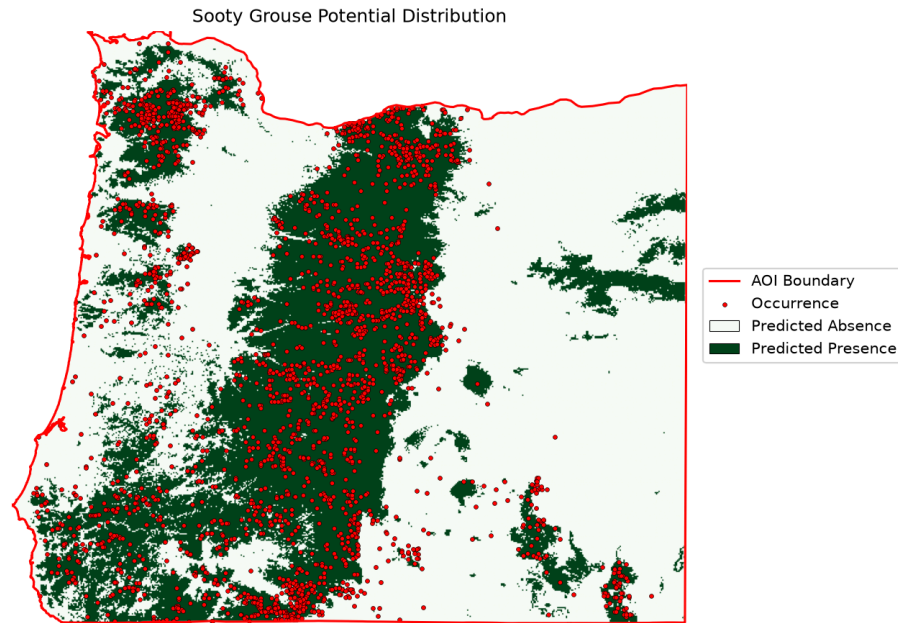


Figure 11. Predicted Sooty Grouse potential distribution across Oregon, based on majority-vote binary classification across 5 cross-validation folds.

Discussion. This model identified total precipitation and elevation as the strongest predictors of Sooty Grouse habitat suitability, consistent with the species' known reliance on montane conifer forest and elevational movement between breeding and wintering habitat (Zwickel & Bendell, 2024; Ahlborn, 2008).

Elevation was retained despite a comparatively high VIF (8.77) and moderate correlation with temperature ($r = -0.73$) and mixed forest cover ($r = -0.60$), on the basis that it integrates several ecologically relevant gradients not fully captured by any single correlated variable. This reflects a broader principle in SDM work: multicollinearity thresholds should inform, not replace, ecologically grounded variable selection.

Several design choices departed from the Google Earth Engine tutorial (Yu, 2024) this project was based on. Pseudo-absence points were generated using buffer exclusion rather than environmental clustering, since k-means clustering exceeded this project's Earth Engine computational limits. Occurrence data also came from GBIF citizen science records rather than structured surveys, introducing potential observer effort bias (e.g., overrepresentation near roads and population centers) relative to the systematically surveyed datasets used in comparable grouse studies. Finally, the evaluation used 5 cross-validation folds rather than a larger number, balancing robustness against computational cost. The low variance across folds (± 0.008) suggests a stable performance estimate despite the reduced count.

Despite these limitations, the resulting maps provide a baseline, state-scale characterization of Sooty Grouse habitat suitability that could inform survey and conservation planning in Oregon. Future work could incorporate occurrence data from neighboring states, and explore environmentally-stratified pseudo-absence sampling at a more computationally tractable scale.

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